

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

BASIC AUTOMOBILE ENGINEERING**MODEL QUESTION PAPER**

Time : 3 Hours

Max. Marks : 75

PART A**I. Answer all questions in one word or one sentence****(9 x 1 = 9 Marks)**

1	The system of Automobile which contain battery, charging system, and starting system is termed as a) Electrical system b) Charging system c) Starting system d) Battery system	M 1.02	U
2	Define the term wheel track	M 1.04	R
3	The volume above piston when it is at TDC is known as.....	M2.01	U
4	The compression ratio range of a petrol engine is.....	M 2.01	U
5	List any three materials used for the construction of cylinder head	M 3.01	R
6	What is the function of compression ring in a piston	M 3.01	R
7	The firing order of a 4 cylinder engine is-----	M 4.01	U
8	Recall the function of a vibration damper in an engine	M 4.01	R
9	What is the material used for Inlet valve of an engine	M 4.03	R

PART B**II. Answer any eight questions from the following. Each question carries 3 marks****(8 x 3= 24 Marks)**

1	Define the following terms (a) LMV (b) HTV	M 1.02	R
2	Compare gross weight and kerb weight of a vehicle	M 1.04	U
3	List the major components of an engine cooling system.	M 1.03	R
4	Explain the relationship between Indicated power and frictional power of an Internal Combustion. engine	M 2.01	U
5	Explain the difference between Internal Combustion. engine and External Combustion. engine	M 2.01	U
6	Outline the functions of any three gaskets used in engines .	M 3.02	U
7	List any three expansion control methods in pistons.	M 3.03	R
8	How do a flywheel is constructed?	M4.01	R
9	Define the function of camshaft in an engine	M 4.02	R
10	Compare the functions of (a) valve seat and (b) valve guide	M 4.03	U

PART C

III. Answer all questions. Each question carries seven marks

(6x 7 = 42 Marks)

1	Explain plant location and latest products of any one Indian automotive manufacturer.	M 1.01	U
	OR		

2	Demonstrate chassis layout of a four wheeler and prepare a component list.	M 1.03	U
3	Illustrate the working of a two stroke petrol engine.	M 2.04	U
	OR		
4	Illustrate the working of a four stroke diesel engine.	M 2.05	U
5	Identify any seven differences between petrol engine and diesel engine	M 2.01	A
	OR		
6	Identify any seven differences between two stroke engine and four stroke engine	M 2.04	A
7	Illustrate the constructional details of a dry liner	M 3.03	U
	OR		
8	Explain the constructional features of a piston	M 3.03	U
9	Summarize the parts of a crankshaft with a neat sketch	M 3.03	U
	OR		
10	Illustrate the parts of a typical connecting rod .	M 3.03	U
11	Explain overhead valve mechanism with the aid of a diagram.	M 4.03	U
	OR		
12	Summarize the features of a poppet valve with a neat sketch	M4.03	U